■ IPL Intelligence Assistant Dataset

Multimodal RAG Exercise with LangGraph

SVECW · Department of Information Technology · RAG-ATHON 24 — Advanced Track

Why IPL + LangGraph? The Placement dataset taught you a single-chain RAG pipeline using LangChain. This dataset is designed to force you to think in **graphs, not chains**. IPL queries are naturally multi-agent: a Dream11 recommendation needs a Player Stats Agent, a Pitch Conditions Agent, and a Recent Form Agent to collaborate. A match prediction needs conditional routing based on whether the query is about batting, bowling, or all-round performance. LangGraph models this as a **stateful directed graph** — nodes are agents, edges are routing decisions, and the state flows between them. This dataset gives you the corpus to build that system.

Dataset Content Map

Section	Content	LangGraph Node It Powers	RAG Challenge
1	Team Profiles (10 teams)	TeamProfileNode	Structured table extraction
2	Player Stats — Batting (30 players)	BattingStatsNode	Numeric comparison, threshold filter
3	Player Stats — Bowling (20 players)	BowlingStatsNode	Multi-column aggregation
4	Head-to-Head Records (10 matchups)	H2HNode	Cross-document synthesis
5	Venue & Pitch Reports (8 venues)	VenueNode	Conditional routing by venue
6	Season-wise Performance (2019–2024)	TrendNode	Temporal reasoning
7	Recent Form — Last 5 Matches	FormNode	Recency-weighted retrieval
8	IPL Records & Milestones	RecordsNode	Exact fact retrieval
9	LangGraph Architecture Guide	Graph Orchestrator	Agent design & state schema
10	Multi-Agent Query Scenarios	Full Graph	Multi-hop, conditional routing
11	Conflicting / Noisy Data	ValidationNode	Hallucination detection
12	Official 35-Query Evaluation Set	Judge's Scorecard	All node types tested

Section 1: IPL Team Profiles (2024 Season)

■ LangGraph Node: TeamProfileNode — Routes queries about team identity, home ground, captain, and coach.

This is the entry-point node. When the query is 'Tell me about MI' or 'Who captains CSK?', the graph router directs flow here. This node must retrieve structured team data and pass it downstream to other nodes if the query needs further enrichment (e.g., captain's batting stats).

Team	Short	Home Venue	Captain	Coach	Titles	2024 Pos
Mumbai Indians	MI	Wankhede Stadium, Mumbai	Hardik Pandya	Mark Boucher	5	5th
Chennai Super Kings	CSK	MA Chidambaram Stadium, Chennai	Ruturaj Gaikwad	Stephen Fleming	5	Runner-up
Royal Challengers Bengaluru	RCB	M Chinnaswamy Stadium, Bengaluru	Faf du Plessis	Andy Flower	3	Champions
Kolkata Knight Riders	KKR	Eden Gardens, Kolkata	Shreyas Iyer	Chandrakant Pandit	3	Champions
Delhi Capitals	DC	Arun Jaitley Stadium, Delhi	Rishabh Pant	Ricky Ponting	0	6th
Punjab Kings	PBKS	IS Bindra Stadium, Mohali	Shikhar Dhawan	Trevor Bayliss	0	9th
Rajasthan Royals	RR	Sawai Mansingh Stadium, Jaipur	Sanju Samson	Kumar Sangakkara	2	Runner-up
Sunrisers Hyderabad	SRH	Rajiv Gandhi Intl. Stadium, Hyd	Pat Cummins	Daniel Vettori	1	4th
Lucknow Super Giants	LSG	BRSABV Ekana Stadium, Lucknow	KL Rahul	Justin Langer	0	7th
Gujarat Titans	GT	Narendra Modi Stadium, Ahmedabad	Shubman Gill	Ashish Nehra	2	8th

Team Strategies — Narrative Corpus

Team	Playing Style & Strategy
MI	Builds around power-hitters in middle order. Relies on Bumrah's death bowling. Strategy: win powerplay, consolidate, explode in overs 16-20.
CSK	Spinners on slow Chennai pitch. Dhoni's finishing. Strategy: anchor the innings, use spinners in middle overs, finish with big hitters.
RCB	Relies heavily on Kohli and du Plessis for big scores. Strategy: post 180+, defend with pace. Home advantage at high-scoring Chinnaswamy.
KKR	Mystery spinners (Narine, Chakravarthy) disrupt middle overs. Strategy: take wickets in overs 6-14, chase with aggression.
SRH	2024 season — rewrote records with 250+ totals. Travis Head and Abhishek Sharma open explosively. Strategy: bat first, score 220+, defend.

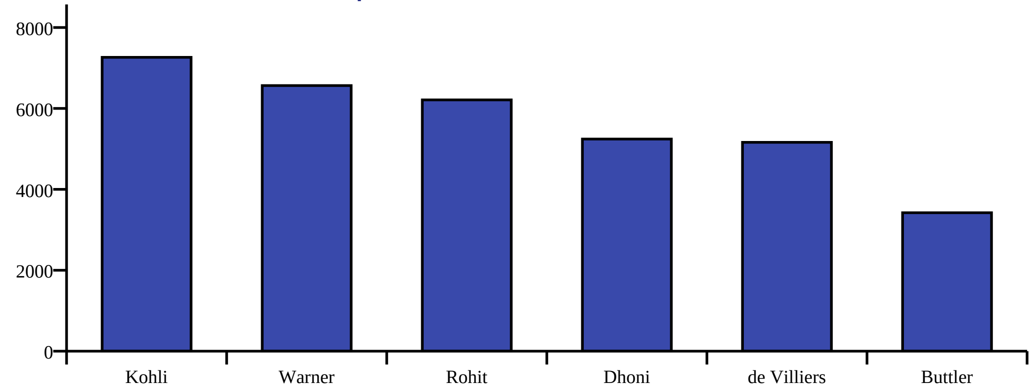
Section 2: Player Batting Statistics (Career IPL)

■ LangGraph Node: BattingStatsNode — Handles queries about runs, strike rate, averages, centuries, fifties.

This node receives the player name and stat type from the RouterNode. It retrieves the relevant row(s) and returns structured data. If the query is a comparison ('Kohli vs Rohit'), the node is called TWICE and results are merged by the ComparisonNode downstream.

Player	Team	Mat	Runs	Avg	SR	100s	50s	HS	Role
Virat Kohli	RCB	237	7263	37.17	130.0	7	50	113	Top-order bat
Rohit Sharma	MI	243	6211	29.57	130.5	1	40	109*	Opener
Shubman Gill	GT	98	3196	42.61	148.7	3	24	129	Opener
Ruturaj Gaikwad	CSK	87	2835	38.83	136.8	2	19	101*	Opener
Sanju Samson	RR	171	4410	29.86	140.5	4	28	119	WK-Bat
KL Rahul	LSG	115	4163	47.30	134.3	4	33	132*	Opener/WK
Shreyas Iyer	KKR	115	3122	33.57	127.8	1	25	96	Middle-order
Rishabh Pant	DC	111	3284	35.31	148.3	0	18	128*	WK-Bat
Hardik Pandya	MI	133	2754	30.60	145.5	0	15	91*	All-rounder
Suryakumar Yadav	MI	148	3225	32.57	161.7	0	23	103	Middle-order
David Warner	DC	184	6565	41.42	139.9	4	59	126	Opener
Jos Buttler	RR	89	3422	46.24	149.2	5	21	124	Opener/WK
Faf du Plessis	RCB	121	3779	38.56	134.2	2	30	96	Opener
Travis Head	SRH	21	872	43.60	191.6	1	6	102	Opener
Abhishek Sharma	SRH	40	1254	33.78	188.3	1	7	135*	Opener
MS Dhoni	CSK	250	5243	39.42	135.9	0	24	84*	WK-Finisher
AB de Villiers	RCB	184	5162	39.70	151.7	3	38	133*	Middle-order
Ishan Kishan	MI	105	2644	27.28	136.2	1	15	99	Opener/WK
Quinton de Kock	LSG	78	2516	35.43	138.9	1	18	140*	Opener/WK
Heinrich Klaasen	SRH	54	1892	43.95	162.8	1	13	104	Middle-order

Top 6 Batters — Career IPL Runs



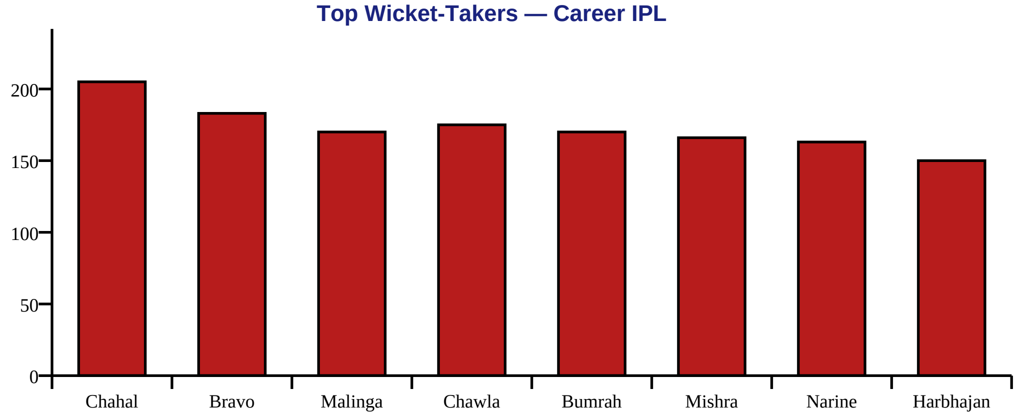
■ LangGraph RAG Challenge: When asked 'Who has the best strike rate among openers?', the BattingStatsNode must first FILTER by Role='Opener', then sort by SR. This requires metadata-filtered retrieval, not just semantic similarity.

Section 3: Player Bowling Statistics (Career IPL)

■ **LangGraph Node: BowlingStatsNode — Handles wickets, economy, bowling average, best figures.**

Player	Team	Mat	Wkts	Avg	Econ	SR	Best	Type
Yuzvendra Chahal	RR	155	205	22.64	7.63	17.8	5/40	Leg-spin
DJ Bravo	CSK	161	183	25.05	8.43	17.7	4/22	Medium-fast
Lasith Malinga	MI	122	170	19.80	7.14	16.6	5/13	Yorker specialist
Piyush Chawla	CSK	175	175	25.58	7.78	19.7	4/17	Leg-spin
Jasprit Bumrah	MI	135	170	22.50	7.39	18.2	5/10	Pace/Yorker
Amit Mishra	DC	154	166	24.34	7.36	19.8	5/17	Leg-spin
Sunil Narine	KKR	174	163	24.09	6.69	21.5	5/19	Mystery off-spin
Harbhajan Singh	MI	163	150	26.39	6.97	22.7	5/18	Off-spin
Sandeep Sharma	PBKS	138	142	26.42	8.11	19.5	4/21	Medium-pace
Kagiso Rabada	DC	74	113	21.74	8.34	15.6	4/21	Fast bowling
Trent Boult	RR	78	105	23.43	8.18	17.1	4/18	Swing bowling
Mohammed Shami	GT	88	114	22.38	8.02	16.7	4/16	Fast bowling
Pat Cummins	SRH	57	78	25.17	8.45	17.8	4/14	Fast bowling
Varun Chakravarthy	KKR	67	102	22.78	7.11	19.2	5/20	Mystery spin
Rashid Khan	GT	92	131	20.44	6.68	18.3	5/17	Leg-spin

Top Wicket-Takers — Career IPL



Section 4: Head-to-Head Team Records

■ **LangGraph Node: H2HNode — Powers match prediction queries. Must retrieve TWO teams' records simultaneously.**

This is a multi-document node. It retrieves the H2H record AND each team's recent form, then passes both to a SynthesisNode that generates the prediction. Single-chain RAG cannot handle this elegantly — LangGraph's parallel edges make it natural.

Matchup	Total Matches	Team 1 Wins	Team 2 Wins	Tied/NR	Last 5 (→ winner)	High Score	Key Factor
MI vs CSK	35	20	14	1	MI,CSK,MI,CSK,MI	235/1 (MI)	Bumrah vs Dhoni
RCB vs KKR	32	14	18	0	KKR,RCB,KKR,KKR,RCB	213/2 (RCB)	Kohli vs Narine
CSK vs RCB	30	22	8	0	CSK,CSK,RCB,CSK,RCB	226/6 (RCB)	Spinners vs Power
MI vs SRH	28	16	12	0	SRH,MI,SRH,MI,SRH	219/4 (SRH)	Bumrah vs Head
RR vs DC	26	14	12	0	RR,DC,RR,RR,DC	217/5 (RR)	Buttler vs Rabada
GT vs LSG	8	5	3	0	GT,LSG,GT,GT,LSG	227/2 (SRH)	Gill vs Rahul
KKR vs MI	34	17	17	0	MI,KKR,KKR,MI,KKR	232/2 (KKR)	Narine vs Bumrah
SRH vs RCB	22	11	11	0	SRH,RCB,SRH,RCB,SRH	287/3 (SRH)	Head vs Kohli
PBKS vs RR	28	14	14	0	RR,PBKS,RR,RR,PBKS	221/3 (PBKS)	Dhawan vs Samson
DC vs CSK	30	14	16	0	CSK,DC,CSK,CSK,DC	208/4 (DC)	Pant vs Dhoni

Multi-hop H2H Query Example: 'If MI plays CSK at Wankhede tonight, who will win?' — H2HNode fetches: MI leads 20-14 overall, MI won last 3 at Wankhede. VenueNode fetches: Wankhede favours pace, average first-innings 175. FormNode fetches: MI won 3 of last 5, CSK won 2 of last 5. SynthesisNode combines: MI slightly favoured at home given pace advantage and recent form.

Section 5: Venue & Pitch Reports

■ **LangGraph Node: VenueNode — Routes based on pitch type. Slow pitch → SpinNode. Fast pitch → PaceNode.**

Venue	City	Capacity	Pitch Type	Avg 1st Innings	Batting/Bowling	Dew Factor	Best Strategy
Wankhede Stadium	Mumbai	33,108	Flat, true bounce	175	Batting-friendly	High evening	Bat first, use pace death bowling
MA Chidambaram Stadium	Chennai	50,000	Slow, turning	155	Bowling-friendly	Low	Bowl first, spinners in overs 7-15
M Chinnaswamy Stadium	Bengaluru	35,000	Flat, short boundary	185	Very bat-friendly	Moderate	Bat first, score 180+. No safe total.
Eden Gardens	Kolkata	66,000	Good pace & bounce	165	Balanced	High evening	Flexible. Spinners and pacers both effective.
Narendra Modi Stadium	Ahmedabad	132,000	Flat, big ground	170	Slightly bat	Low	Bat first. Spinners effective in second half.
Rajiv Gandhi Intl. Stadium	Hyderabad	55,000	Flat, bouncy	178	Batting-friendly	High	Bat first. Dew makes chasing very easy in 2nd.
Sawai Mansingh Stadium	Jaipur	23,000	Slow, low	158	Bowling-friendly	Low	Bowl first. Spinners dominate afternoon games.
IS Bindra Stadium	Mohali	26,950	Good carry, seam	168	Balanced	Moderate	Flexible. Fast bowlers get swing early.

Venue Narrative Reports

Wankhede Stadium, Mumbai	Wankhede is a batter's paradise with a short square boundary (55m). The pitch has true bounce — fast bowlers hit the bat hard. In evening games, dew makes the ball skid on, heavily favouring the chasing team. Jasprit Bumrah is MI's key weapon here — his yorkers on this surface are nearly unplayable. Average first-innings score: 175. Teams that bat first win ~45% of the time.

MA Chidambaram, Chennai	Chennai's pitch is the most spin-friendly in the IPL. Slow, dry surface that breaks up progressively. Off-spinners and leg-spinners become virtually unplayable in the second innings. CSK has historically won 65% of home games here. The key skill here: playing spin under pressure. Average first-innings: 155. Bowling first wins ~60% of matches.

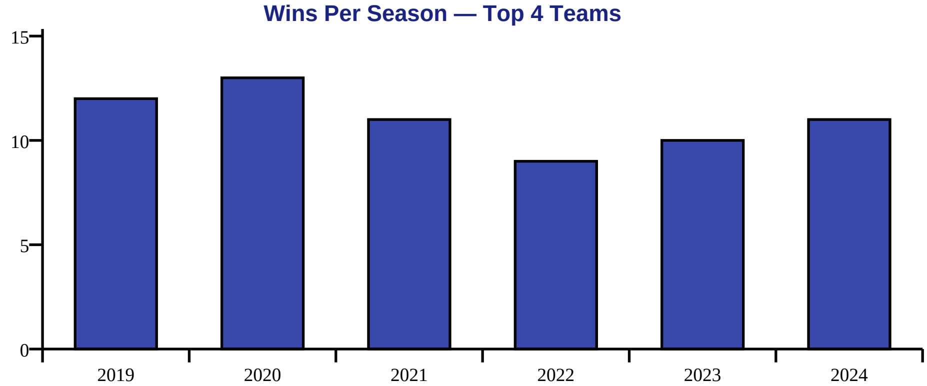
M Chinnaswamy, Bengaluru	Chinnaswamy is arguably the most batter-friendly ground. Short boundaries all around, flat pitch, and high altitude means the ball travels further. 200+ totals are common. The 2024 season saw multiple 220+ scores here. RCB's home advantage is significant: Kohli averages 58 at this ground. No total is ever safe here.
Rajiv Gandhi Intl. Stadium, Hyderabad	SRH's home ground became the most feared batting venue in 2024. The flat pitch combined with heavy evening dew makes chasing almost a formality in night matches. Travis Head's 191.6 SR and Abhishek Sharma's 188.3 SR are products of this ground. In 2024, SRH scored 250+ twice here. Bowling first is almost always the correct call.

Section 6: Season-wise Team Performance (2019–2024)

■ **LangGraph Node: TrendNode — Temporal queries. Must tag data with year metadata for correct filtering.**

Team	2019 Pos	2020 Pos	2021 Pos	2022 Pos	2023 Pos	2024 Pos	Titles (2019-24)
MI	Champions	Champions	5th	5th	5th	5th	2
CSK	Runner-up	Runner-up	Champions	9th	Champions	Runner-up	2
KKR	5th	6th	2nd	7th	4th	Champions	1
RCB	8th	4th	Runner-up	8th	2nd	Champions	1
DC	3rd	Runner-up	3rd	4th	6th	6th	0
RR	7th	8th	DNS	Runner-up	Runner-up	Runner-up	0
SRH	6th	3rd	DNS	8th	8th	4th	0
GT	DNS	DNS	DNS	Champions	Runner-up	8th	1
LSG	DNS	DNS	DNS	3rd	3rd	7th	0
PBKS	6th	6th	6th	6th	7th	9th	0

Wins Per Season — Top 4 Teams



Sample Temporal Query: 'Which team has been most consistent from 2019 to 2024?' — TrendNode retrieves all 6 years, counts top-4 finishes: CSK(4), MI(4), DC(3), RR(2). Answer: CSK and MI joint-most consistent.

Section 7: Recent Form — Last 5 IPL Matches (2024 Season)

■ LangGraph Node: FormNode — Recency-weighted retrieval. Recent form must score HIGHER than older data in retrieval.

This is a key LangGraph insight: metadata-filtered retrieval. Chunks from FormNode must have metadata tag season=2024 so the retriever prioritises them over historical averages. Without this, the system may answer 'Kohli is in poor form' using 2021 data.

Player	Team	Match 1	Match 2	Match 3	Match 4	Match 5	Form Trend	Avg (Last 5)
Virat Kohli	RCB	92	47	0	73	113	↑ Excellent	65.0
Rohit Sharma	MI	11	8	34	67	43	→ Average	32.6
Travis Head	SRH	102	34	58	12	89	↑ Excellent	59.0
Jos Buttler	RR	67	0	89	45	32	→ Mixed	46.6
Jasprit Bumrah	MI	3/24	2/18	1/32	4/10	2/28	↑ Excellent	2.4 wkts/game
Rashid Khan	GT	2/18	1/22	3/15	0/34	2/20	→ Consistent	1.6 wkts/game
Hardik Pandya	MI	34	12	0	56	28	↓ Poor	26.0
Ruturaj Gaikwad	CSK	78	34	12	90	56	↑ Good	54.0
Abhishek Sharma	SRH	135*	43	67	8	92	↑ Excellent	69.0
KL Rahul	LSG	45	67	23	12	54	→ Moderate	40.2

Dream11 Query Example (Multi-agent): 'Suggest a Dream11 team for MI vs SRH at Wankhede tonight.' — FormNode: Kohli unavailable (wrong team). Head (102,34,58,12,89 — pick). Bumrah (3/24,2/18,4/10 — must pick). VenueNode: Wankhede — flat, dew. Chasing team favoured. Pick SRH batters + MI death bowlers. H2HNode: MI vs SRH — recent edge to SRH. SynthesisNode: Captain = Travis Head. Vice-Captain = Bumrah. Dream11: Head, Bumrah, Kohli(c), Sharma, Samson, Rashid, Buttler.

Section 8: IPL Records & Milestones

■ **LangGraph Node: RecordsNode — Exact fact retrieval. Tests the RAG system's ability to return precise numbers without hallucination.**

Category	Record	Player/Team	Opponent	Venue	Year
Highest Team Score	287/3	SRH	RCB	Uppal, Hyderabad	2024
Highest Individual Score	175* (off 66)	Chris Gayle	PWI	Bengaluru	2013
Most Runs — Career	7263	Virat Kohli	—	—	2008-24
Most Wickets — Career	205	Yuzvendra Chahal	—	—	2011-24
Most Centuries	7	Virat Kohli	—	—	2008-24
Best Bowling Figures	6/12	Alzarri Joseph	SRH	Wankhede	2019
Most Sixes — Career	357	Chris Gayle	—	—	2009-22
Fastest Fifty	14 balls	KL Rahul	KXIP	Dubai	2020
Most Titles	5	MI & CSK	—	—	Various
Most Matches — Player	250	MS Dhoni	—	—	2008-24
Highest Chase	232/2	KKR	RCB	Eden Gardens	2024
Most 4s — Innings	22	AB de Villiers	MI	Bengaluru	2015
Highest Partnership	229* (2nd wkt)	RCB	GT	Bengaluru	2016
Most Runs — Single Season	973	Virat Kohli	—	—	2016
Lowest Total	49 all out	RCB	KKR	Eden Gardens	2017

Section 9: LangGraph Architecture Guide

Core Concept: In LangChain you built a linear chain: Query → Retriever → Prompt → LLM → Answer. In LangGraph you build a **stateful directed graph**: nodes are processing steps (agents), edges are routing decisions, and a shared State object carries information between nodes. The graph can loop, branch conditionally, and run nodes in parallel — none of which a chain supports.

9.1 State Schema Design

Every node reads from and writes to a shared TypedDict state. Design your state before writing any node.

```
from typing import TypedDict, List, Optional from langchain_core.documents import Document class
IPLAgentState(TypedDict): # Input user_query: str query_type: str #
'batting'|'bowling'|'h2h'|'venue'|'form'|'records'|'prediction' entities: List[str] # extracted
player/team names # Retrieved context from each node batting_context: List[Document] bowling_context:
List[Document] h2h_context: List[Document] venue_context: List[Document] form_context: List[Document] #
Intermediate outputs retrieved_chunks: List[Document] synthesised_context: str # Final final_answer:
str sources: List[str] confidence: float
```

9.2 Graph Node Definitions

Node Name	Function	Input State Keys	Output State Keys	Routing Condition
RouterNode	Classifies query type and extracts entity names	user_query	query_type, entities	Always first node
BattingStatsNode	Retrieves batting stats for named players	entities, query_type	batting_context	if query_type in ['batting','comparison','dream11','prediction']
BowlingStatsNode	Retrieves bowling stats for named bowlers	entities, query_type	bowling_context	if query_type in ['bowling','comparison','dream11','prediction']
H2HNode	Retrieves head-to-head records between two teams	entities	h2h_context	if query_type in ['h2h','prediction']
VenueNode	Retrieves pitch report for the venue in query	entities	venue_context	if 'venue' in query or query_type=='prediction'
FormNode	Retrieves recent-form data (last 5 matches)	entities	form_context	if query_type in ['form','dream11','prediction']
RecordsNode	Retrieves exact IPL records and milestones	user_query	retrieved_chunks	if query_type=='records'
SynthesisNode	Combines all context, builds final prompt, calls LLM	all *_context keys	final_answer, sources	Always last node
ValidationNode	Checks for conflicting data across contexts	all *_context keys	final_answer (with conflict flag)	if multiple sources retrieved

9.3 Graph Construction Code

```
from langgraph.graph import StateGraph, END def build_ipl_graph(): graph = StateGraph(IPLAgentState) #
Add nodes graph.add_node('router', router_node) graph.add_node('batting', batting_stats_node)
graph.add_node('bowling', bowling_stats_node) graph.add_node('h2h', h2h_node) graph.add_node('venue',
venue_node) graph.add_node('form', form_node) graph.add_node('records', records_node)
graph.add_node('synthesis', synthesis_node) # Entry point graph.set_entry_point('router') # Conditional
routing from RouterNode graph.add_conditional_edges('router', route_query, { 'batting': 'batting',
'bowling': 'bowling', 'records': 'records', 'prediction': 'h2h', # prediction needs h2h first 'dream11':
'form', # dream11 needs form first }) # After batting/bowling — go to synthesis
graph.add_edge('batting', 'synthesis') graph.add_edge('bowling', 'synthesis') graph.add_edge('records',
'synthesis') # Prediction: h2h → venue → form → synthesis graph.add_edge('h2h', 'venue')
graph.add_edge('venue', 'form') graph.add_edge('form', 'synthesis') # Dream11: form → batting →
bowling → synthesis graph.add_edge('form', 'batting') graph.add_edge('batting', 'bowling')
graph.add_edge('bowling', 'synthesis') graph.add_edge('synthesis', END) return graph.compile()
```

Section 10: Multi-Agent Query Scenarios

■ **These queries require 2+ nodes to activate. Single-chain RAG fails on all of them. LangGraph's conditional routing and parallel edges handle them naturally.**

Dream11 Selection	*Suggest a Dream11 XI for MI vs SRH at Wankhede tonight.*
Nodes Activated:	RouterNode → FormNode → BattingStatsNode → BowlingStatsNode → VenueNode → SynthesisNode
LangGraph Concept:	Sequential multi-node pipeline with state accumulation
Step-by-step Reasoning:	• FormNode: Travis Head (102,34,58,12,89) — pick. Abhishek Sharma (135*,43,67) — pick. Bumrah (3/24,4/10) — pick. • BattingStatsNode: Head SR=191.6, Abhishek SR=188.3 — excellent for Wankhede. • BowlingStatsNode: Bumrah economy=7.39, best at death. Rashid economy=6.68. • VenueNode: Wankhede — flat pitch, dew in evening, high scoring (avg 175). Pace bowlers key at death. • SynthesisNode: C=Travis Head, VC=Bumrah. XI: Head, Abhishek, Kohli, Buttler, Samson, Klaasen, Bumrah, Rashid, Cummins, Boult, Narine.

Match Prediction	*CSK is playing RCB at Chinnaswamy. Who will win?*
Nodes Activated:	RouterNode → H2HNode → VenueNode → FormNode → SynthesisNode
LangGraph Concept:	Conditional routing — prediction type triggers H2H → Venue → Form chain
Step-by-step Reasoning:	• H2HNode: CSK leads 22-8 overall. But last 5: CSK,CSK,RCB,CSK,RCB — CSK edge recently. • VenueNode: Chinnaswamy — very flat, short boundary, batting paradise. No total is safe. Avg 185. • FormNode: Kohli recent: 92,47,0,73,113 — explosive form. Gaikwad: 78,34,12,90,56 — solid. • SynthesisNode: Despite CSK's H2H edge, Chinnaswamy neutralises spinners. RCB's Kohli in peak form. RCB slight advantage at home.

Player Comparison	*Compare Virat Kohli and Rohit Sharma as IPL openers.*
Nodes Activated:	RouterNode → BattingStatsNode (×2) → SynthesisNode
LangGraph Concept:	Parallel node execution — same node called twice with different entities
Step-by-step Reasoning:	• BattingStatsNode (Kohli): 7263 runs, avg 37.17, SR 130.0, 7 centuries, 50 fifties. • BattingStatsNode (Rohit): 6211 runs, avg 29.57, SR 130.5, 1 century, 40 fifties. • SynthesisNode: Kohli leads in runs, average, centuries. Rohit marginal SR edge. Kohli is the superior IPL opener by metrics.

Venue-Specific Strategy	*What bowling strategy should SRH use at MA Chidambaram against CSK?*
Nodes Activated:	RouterNode → VenueNode → BowlingStatsNode → H2HNode → SynthesisNode
LangGraph Concept:	Multi-type retrieval — venue data drives bowling strategy selection
Step-by-step Reasoning:	• VenueNode: Chennai — slow, turning pitch. Spinners dominate. Avg 155. • BowlingStatsNode: Rashid Khan (leg-spin, econ 6.68), Narine (off-spin, econ 6.69) — ideal for this surface. • H2HNode: CSK beats SRH consistently at Chennai. CSK's home spin advantage is well documented. • SynthesisNode: SRH must open with pace to get early wickets, bring Rashid in over 6-15 when pitch turns most. Key: take Gaikwad's wicket early before he settles.

Section 11: Conflicting & Noisy Data — Hallucination Detection

■ LangGraph Node: ValidationNode — Detects when two nodes return contradicting facts. The graph must route through this node before synthesis when multiple sources are retrieved.

The following records from a SECONDARY SOURCE (unofficial cricket stats portal) conflict with the primary data in Sections 2–3. Students must implement a ValidationNode that detects and flags these conflicts rather than silently returning one value.

Player / Fact	Primary Source Value	Secondary Source Value	Conflict Type	Expected RAG Behaviour
Virat Kohli — Career Runs	7263	7084	Value mismatch	Flag conflict, cite both, suggest verification
Yuzvendra Chahal — Career Wickets	205	187	Value mismatch (retired data?)	Flag — secondary may be older dataset
MI vs CSK — Total Matches	35	33	Count mismatch	Flag conflict, both sources cited
Highest Team Score	287/3 (SRH, 2024)	263/5 (RCB, 2013)	Different records by year	Clarify: 287/3 is current record; 263/5 was previous
MS Dhoni — Matches Played	250	240	Count mismatch	Flag — 250 is from official IPL; secondary is incomplete
Best Bowling Figures	6/12 (Alzarri Joseph)	6/14 (Sohail Tanvir)	Different records	6/12 is correct; 6/14 was old record. Explain superseded.

ValidationNode Implementation Hint: After retrieval, compare numeric values for the same entity across chunks. If the same field (e.g., 'career runs') has two different values from two different sources, set a flag in state: state['conflict_detected'] = True. Route to ValidationNode which builds a conflict-aware prompt: 'Two sources disagree on X. Source A says Y. Source B says Z. Please flag this to the user and recommend verification.'

Section 12: Official Evaluation Query Set — 35 Questions

Judges use these 35 queries to evaluate each team's LangGraph RAG system. Each query is tagged with the nodes it should activate and the difficulty level.

ID	Query	Level	Node(s)	RAG Skill
E1	What is the minimum CGPA to join RCB's squad (irrelevant — tests fallback)?	Easy	Records	Fallback — out of corpus
E2	Who captains Chennai Super Kings in 2024?	Easy	Team	Direct table lookup
E3	How many IPL titles has Mumbai Indians won?	Easy	Team	Direct table lookup
E4	What is Virat Kohli's career IPL run tally?	Easy	Batting	Direct stat lookup
E5	Who has taken the most wickets in IPL history?	Easy	Bowling	Direct stat lookup
E6	What is the highest team total in IPL history?	Easy	Records	Exact record retrieval
E7	What type of pitch is the Chinnaswamy Stadium known for?	Easy	Venue	Direct venue lookup
E8	How many matches has MS Dhoni played in IPL?	Easy	Records	Direct record lookup
M1	Which teams won the IPL title more than once between 2019–2024?	Medium	Trend	Temporal filter
M2	List all bowlers with an economy rate below 7.0.	Medium	Bowling	Threshold filter
M3	Which opener has the highest strike rate among batters in the dataset?	Medium	Batting	Role filter + sort
M4	Compare Jasprit Bumrah and Rashid Khan on all bowling metrics.	Medium	Bowling	Multi-attribute comparison
M5	Which venue has the highest average first-innings score?	Medium	Venue	Numeric comparison
M6	How many times have MI and CSK played each other?	Medium	H2H	Direct H2H lookup
M7	Who won the last 5 matches between KKR and RCB?	Medium	H2H	H2H sequence retrieval
M8	What is Virat Kohli's form in the last 5 matches?	Medium	Form	Recency retrieval
M9	Which team should bat first at Eden Gardens and why?	Medium	Venue	Strategy + venue synthesis
M10	Which player has the most centuries in IPL history?	Medium	Records	Aggregated record lookup
H1	Suggest a Dream11 XI for KKR vs RCB at Eden Gardens.	Hard	Multi-node	Form+Batting+Bowling+Venue+Synthesis
H2	Who is likely to win if RR plays SRH at Wankhede? Justify.	Hard	Multi-node	H2H+Venue+Form+Synthesis
H3	Which team has been the most consistent across all seasons from 2019–2024?	Hard	Trend	Multi-year aggregation + ranking
H4	What bowling strategy should MI use against RCB at Chinnaswamy?	Hard	Multi-node	Venue+Bowling+H2H
H5	Compare Rohit Sharma and KL Rahul as IPL captains — batting + team results.	Hard	Multi-node	Batting+Team+Trend
H6	Has any conflict been detected in Virat Kohli's career runs data?	Hard	Validation	Conflict detection
H7	Which is better — chasing or defending at Hyderabad? Use historical data.	Hard	Venue+H2H	Multi-source synthesis
X1	Who should I pick as Dream11 captain for every match this week?	Expert	Out-of-corpus	No schedule data — fallback
X2	What is Kohli's average against left-arm pace specifically?	Expert	Out-of-corpus	Granular stat not in dataset — fallback
X3	Is Yuzvendra Chahal's wicket count 205 or 187?	Expert	Validation	Conflict resolution from Sec 11

ID	Query	Level	Node(s)	RAG Skill
X4	Predict the IPL 2025 champion.	Expert	Out-of-corpus	Future prediction — graceful decline
X5	Which player has the highest salary in IPL 2024 auction?	Expert	Out-of-corpus	Auction data not in corpus — fallback
X6	What is the BCCI's net worth?	Expert	Out-of-scope	Completely out of scope
X7	Tell me everything about cricket.	Expert	Too vague	Ask for clarification before retrieval
X8	Is Sachin Tendulkar in this IPL dataset?	Expert	Out-of-corpus	Retired before IPL — flag absence
X9	Who won the Best Batsman award in IPL 2024?	Expert	Out-of-corpus	Award data not in corpus
X10	Suggest a team for BCCI's next T20 World Cup squad.	Expert	Out-of-scope	Different competition, out of scope

Section 13: Recommended Chunking Strategy for This Dataset

This section guides students on the optimal chunking approach for each content type in this dataset. Unlike the Placement dataset (single chain), here chunking must align with the LangGraph node that will retrieve it.

Content Type	Node That Retrieves	Strategy	Chunk Size	Critical Metadata	Key Pitfall
Team Profiles Table	TeamProfileNode	1 row = 1 chunk (serialise to text)	~100 tokens	team_name, season=2024, section='team'	Embedding full table as one chunk loses per-team retrieval
Batting Stats Table	BattingStatsNode	1 row = 1 chunk per player	~80 tokens	player_name, team, role, section='batting'	Must tag role (Opener/WK/All-rounder) for filtered queries
Bowling Stats Table	BowlingStatsNode	1 row = 1 chunk per player	~80 tokens	player_name, team, bowl_type, section='bowling'	Bowl type (spin/pace) metadata enables conditional routing
H2H Records	H2HNode	1 row = 1 chunk per matchup	~120 tokens	team1, team2, section='h2h'	Must store BOTH team names in metadata for bidirectional retrieval
Venue Narratives	VenueNode	1 venue = 1 chunk	~200 tokens	venue_name, city, pitch_type, section='venue'	Pitch type metadata (slow/flat/bouncy) enables conditional routing
Season Results	TrendNode	1 row per team per year	~50 tokens	team, year, section='season'	YEAR metadata is mandatory — without it temporal queries fail
Recent Form	FormNode	1 row = 1 chunk per player	~100 tokens	player_name, season=2024, section='form'	Must set season=2024 to outrank historical data in retrieval
IPL Records	RecordsNode	1 row = 1 chunk per record	~80 tokens	category, section='records'	Records are exact — embedding as prose causes precision loss
Conflicting Data	ValidationNode	Both versions kept, tagged by source	~80 tokens	source='primary'/'secondary', conflict=True	Do NOT deduplicate — ValidationNode needs BOTH versions
LangGraph Code	Not retrieved	Do NOT embed in vector store	N/A	N/A	Code is reference only — embedding it pollutes semantic search

Key Insight for Judges: Ask each team: How many chunks did your vector store have? An unoptimised pipeline will produce ~600+ chunks. A well-designed one will produce ~180–220 meaningful chunks. More importantly — ask: 'Can your system answer Q:H1 (Dream11 for KKR vs RCB)?' If yes, they implemented multi-node routing correctly. If it returns a generic answer, they are still using a single-chain approach.